



DOW™ Electrical & Telecommunications HFDB-4201 EC K

Product Description

Polyethylene Compound for Power Cable Insulation.

HFDB-4201 EC K is a long-life, cross linkable, low-density, polyethylene insulation compound of high purity developed especially for the insulation of power cables. HFDB-4201 EC K is the European produced HFDB-4201 targeted for medium and lower high voltage cables. HFDB-4201 EC K is equipped with a non-migrating stabilizer providing high thermal stability, long term storage stability, and optimum crosslinking behavior.

Applications:

- HFDB-4201 EC K is a long life cross linkable insulation compound intended for medium and lower high voltage applications.

Specifications:

- The following standards are expected to be met.
- Please consult with Dow TS&D before selecting your project.
 - CENELEC: HD 620 S1/A3, HD632 S2, HD 629 S1
 - DIN VDE: 0207, DIN VDE 0276-620
 - RUS: 50-70 (U-I)
 - IEC: 60840
 - AEIC: CS8
 - ANSI/ICEA: S-94-649, S-97-682, S-93-639/NEMA WC74
 - GB/T 12706

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.921 g/cm ³	0.921 g/cm ³	ISO 1183 ¹
Melt Mass-Flow Rate (130°C/2.16 kg)	0.30 g/10 min	0.30 g/10 min	ISO 1133
Moisture	< 200 ppm	< 200 ppm	Dow Method ²
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength	2900 psi	20.0 MPa	IEC 60811-1-1
Tensile Elongation (Break)	500 %	500 %	IEC 60811-1-1
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Hot Set			IEC 811-2-1 ³
Elongation under Load: 392°F (200°C)	< 100 %	< 100 %	
Permanent Deformation: 392°F (200°C)	< 5.0 %	< 5.0 %	
Aging	Nominal Value (English)	Nominal Value (SI)	Test Method
Change of Tensile Properties After Aging - 10 days			IEC 60811-1-1
302°F (150°C)	< 25 %	< 25 %	
Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Volume Resistivity	> 1.0E+16 ohm·cm	> 1.0E+16 ohm·cm	IEC 60093
Electric Strength	> 760 V/mil	> 30 kV/mm	IEC 60243-1
Dielectric Constant (1 MHz)	< 2.30	< 2.30	IEC 60250
Dissipation Factor (50 Hz)	< 0.00030	< 0.00030	IEC 60250
Additional Information	Nominal Value (English)	Nominal Value (SI)	Test Method
Gottfert Elastograph - Torque	5.3 in·lb	0.60 N·m	ISO 6502
Reaction Speed - t90 (356°F (180°C))	5.0 min	5.0 min	ISO 6502

Storage:

- The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. HFDB-4201 EC K is storage stable even at elevated temperatures. Peroxide may start to migrate at temperatures below 15°C.
- The recommended maximum storage time is 1 year at 40°C and not more than 2 months below 15°C. In general, the material can be used for regular MV applications within two years after production if stored the remaining time between 15 and 30°C. It is recommended that the practice of using the product on a first-in / first-out basis be established.

Packaging:

- HFDB-4201 EC K is delivered in 1000kg octabins.

Extrusion	Nominal Value (English)	Nominal Value (SI)
Melt Temperature	239 to 284 °F	115 to 140 °C

Extrusion Notes

- HFDB-4201 EC K provides excellent surface finish and outstanding output rates over a broad range of conditions very similar to HFDB-4201. For optimum results, melt extrusion temperatures in the range of 115 -140°C are recommended, and preferably 135°C for HV applications.
- Specific recommendations for processing conditions can be determined when the application and processing equipment type are known.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Base Resin

² Karl Fischer titration

³ 0.2 MPa

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